A testing-effort artefact in the most-sensitive aquatic endpoint: structure-based PFAS screening recovers predictive validity but still cannot support clearance

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Abstract

Regulatory hazard screening commonly summarises a substance by the most sensitive effect concentration reported for it. We show that this convention makes the response variable a function of how much the substance has been tested. Across 87 structure-resolved per- and polyfluoroalkyl substances (PFAS) carrying 4,744 freshwater records in the U.S. EPA ECOTOX Knowledgebase, the protective minimum correlated with the logarithm of record count at Spearman $\rho = -0.75$ ($p = 7 \times 10^{-17}$); the most-tested quartile sat 4.9 $\log_{10}$ units below the least-tested. Subsampling every substance to equal depth removed 40–48% of the association at every depth examined, identifying that share as an extreme-order-statistic artefact; the remainder is not attributable to sampling depth alone. The same ordering held across 4,757 chemicals. Holding a published random-forest model fixed and changing only the external target, rank transfer on the same 58 structures rose from $\rho = +0.09$ to $+0.41$ (difference $+0.32$, 95% CI $[+0.08, +0.57]$). Rebuilding training and external sets on an effort-stable acute median restored external validity and narrowed the prediction interval. Clearance nevertheless failed: at any threshold matched to the protective endpoint for stringency, no in-domain PFAS cleared.

Keywords: PFAS, aquatic ecotoxicity, hazard screening, data curation,

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conformal prediction, ECOTOX

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are a structurally diverse class whose environmental persistence has been argued to be sufficient on its own to warrant management at the level of the class [1, 2], and whose global distribution and exposure pathways are now extensively documented [3, 4, 5]. The ecological risk literature for the class has been reviewed in detail [6, 7], and regulatory programmes have set out explicit research and action agendas [8]. Screening programmes for PFAS nonetheless face an inventory far larger than the measured record. The U.S. EPA CompTox Chemicals Dashboard lists tens of thousands of structures [9, 10], while measured freshwater effect concentrations exist for a few hundred. Structure-based models are the obvious way to triage that gap, and quantitative structure–activity relationship (QSAR) models for aquatic toxicity have a long history [11, 12] with current environmental applications combining curated descriptors, fingerprints and machine learning [13, 14, 15]; PFAS-specific models have followed [16], and multi-endpoint hazard prioritisation has been proposed as the operational use for such predictions [17]. Best-practice guidance emphasises chemically independent validation, transparent applicability domains and an explicit endpoint definition [18, 19, 20], and the hazards of reporting internal fit statistics as evidence of predictivity have been set out repeatedly [21].

Structure-based models are now routinely proposed to fill the data gaps that block criteria derivation for emerging contaminants [22, 23, 24]. That a model trained on conventional organic pollutants transfers poorly to PFAS is established [25], and the usual explanation is chemical: perfluorinated chemistry is absent from the training distribution, so the model is extrapolating. We report evidence that a second mechanism, independent of chemistry, contributes substantially to the observed failure.

That mechanism is not unknown qualitatively. The life-cycle assessment literature has long treated the choice between a lowest-value summary and an all-data summary as consequential: USEtox builds its effect factor from an HC50 averaged over all species rather than from the most sensitive species, and a comparison of the two indicator types showed that the choice of data source and endpoint by itself reorders the ecotoxicity ranking of chemicals [26]. A later data-selection sensitivity analysis of 64 persistent and mobile

substances, PFAS among them, reached the same conclusion for characterisation factors [27], and the machine-learning side has noted that "some chemicals have been tested much more often than others" and that the resulting target "rather resembles a probability distribution than a singular ground truth" [28]. The species sensitivity distribution literature is the closest antecedent: it has long been established there that a hazardous concentration estimated from few species is both biased and imprecise relative to one estimated from many, that the number of species tested is itself a determinant of the summary, and that the choice of data and distributional model materially moves the result [29, 30, 31, 32]. What has not been done, to our knowledge, is to measure the effect: to quantify the association between a protective endpoint and testing effort, to separate the part of it that is an extreme-order-statistic artefact from the part that is not, and to show that the residual is large enough to invert the apparent external performance of a published structure-based model. That is what we report.

The mechanism concerns how the response variable is built. Regulatory hazard assessment is conventionally protective: a substance is summarised by the lowest effect concentration reported for it, across species, exposure durations and effect types. The practice is sensible as policy, because the most sensitive observed effect bounds the hazard. As a statistical construction, however, the minimum of a sample is an extreme order statistic, and its expected value decreases as the sample grows. Two substances with identical toxicity will therefore receive different protective values if one has been tested more often than the other. Because testing effort varies by orders of magnitude across the ecotoxicity record — a handful of records for most substances, thousands for PFOA and PFOS — this is not a second-order concern.

If that reasoning is right, three things should follow, and all three are testable. The protective minimum should correlate with the number of available records; equalising the number of records should attenuate the correlation; and a structure-based model, which has no access to testing history, should fail on the protective minimum while succeeding on an endpoint constructed without the confound. We test each in turn, using the complete ECOTOX Knowledgebase rather than a curated subset, and then ask what the corrected picture implies for the decision the screening programme actually wants to make: whether an untested PFAS can be set aside as low concern.

2. Materials and methods

2.1. Data source and record selection

All toxicity data came from the U.S. EPA ECOTOX Knowledgebase ASCII release of 15 September 2026 [33], used in full rather than through a curated derivative such as EnviroTox [34], whose selection rules are appropriate to their own purpose but fix in advance the record counts whose influence is the object of this study. From 725,636 test records we retained tests on aquatic organisms (`organism_habitat` = Water) in freshwater media (`media_type` beginning FW), joined them to 1,250,612 result records, and kept results reporting an acute LC50, EC50 or IC50 or a chronic NOEC, NOEL, LOEC, LOEL or MATC in mass-concentration units convertible to mg/L. Records carrying an inequality qualifier were identified and excluded from the primary analysis rather than treated as point observations, because left-censored values have maximum leverage on the extreme order statistics examined here. Rows identical in test identifier, endpoint type, reported concentration and unit are duplicates of a single observation and were collapsed, because testing effort is measured by counting records and duplicated rows inflate it non-uniformly. This yielded 106,095 acute freshwater records for 4,757 chemicals, and, for the PFAS subset defined below, 15,991 raw rows reducing to 4,744 distinct records from 2,182 tests, carried by 103 CAS registrations that collapse to 87 parent structures under the standardisation of §2.3.

PFAS were identified by CAS match to the PFASSTRUCTV6 list of the CompTox Chemicals Dashboard [10], the same operational definition used in the inventory we later deploy against. Structures were resolved from CAS through PubChem; 4,311 of 4,757 chemicals (90.6%) resolved successfully, and unresolved chemicals were dropped.

2.2. Endpoint constructions

Four summaries of the same records were compared. The *minimum* is the lowest reported value across all acute and chronic strata, which is the protective convention. The *median* is the median over the same records. The *acute median* restricts to acute lethality and concentration-response endpoints before taking the median. The *depth-equalised minimum* draws a fixed number k of records per substance without replacement and averages the resulting minima over repeated draws, isolating the effect of sampling depth while

holding the number of records constant by construction. All are expressed as $y = \log_{10}(T/\mathrm{mg\,L^{-1}})$.

Testing effort was quantified as $\log_{10}$ of the number of usable records for a substance, and its association with each endpoint was measured by Spearman correlation. We emphasise that this is a property of the response variable alone; no model is involved.

2.3. Structure standardisation

Structures were desalted to the fluorine-bearing fragment where present, charge-neutralised with the RDKit standardiser and canonicalised [35, 36]. Desalting to the largest fragment, the more common convention, misassigns salts whose counterion exceeds the active moiety in size, and we therefore prefer the fluorinated fragment explicitly. Substances collapsing to the same parent structure were represented once, and the standardisation was applied before, not after, the per-substance summaries were formed: the records of every registration mapping to one parent were pooled and the endpoint computed once on the pooled set. This matters. Seven of the 103 PFAS CAS registrations collapse to perfluorooctanesulfonate — the free acid and six salts — and their separately assigned protective values span 7.9 $\log_{10}$ units, from -6.00 for the potassium salt to $+1.93$ for the ammonium salt; treated as distinct substances they enter the analysis seven times, at seven different effective testing depths. Collapsing them leaves 87 parent structures. PFAS structures and inventory structures were removed from the broad training set before fitting.

2.4. Model, validation and conformal intervals

The representation was a 2,048-bit ECFP4 fingerprint (Morgan radius 2) [37] and the learner a random-forest regressor with 300 trees, `max_features` $= \sqrt{p}$ and seed 20260620 [38, 39]; these are the settings of the published analysis we re-examine, held fixed so that differences are attributable to the endpoint and not to tuning. Broad performance used five repeats of shuffled five-fold cross-validation over structure-locked rows, with 160 trees per fold. External validation fitted the full broad model once and evaluated it on the PFAS set, with substance-level bootstrap confidence intervals on Spearman ρ (4,000 resamples, except the fixed-model comparison of SI Table S7, which used 3,000).

Normalised split-conformal intervals used the nonconformity score

$$s_i = \frac{|y_i - \hat{y}_i|}{\sigma_i + \beta}, \tag{1}$$

with σ_i the standard deviation across forest trees and β the median calibration σ plus 10^{-6}, and the finite-sample order statistic $\lceil (n+1)(1-\alpha) \rceil$ [40, 41, 42]. Normalising the residual by a model-derived difficulty estimate, rather than using the bare absolute residual, is the standard device for obtaining substance-specific rather than constant-width intervals [43, 44], and the procedure has an established record in QSAR applications [45, 46]. Marginal calibration used a random 80/20 split of the broad set, evaluated on the PFAS set. PFAS-conditional calibration held the broad model fixed and split the PFAS set into disjoint calibration and test halves. Both were repeated 40 times and means reported. The distinction matters because the split-conformal guarantee is marginal over an exchangeable calibration and test pair; conditional validity on a subpopulation is a strictly stronger requirement that marginal calibration does not deliver [47].

2.5. Applicability domain and clearance

The applicability domain was the mean Euclidean distance in ECFP4 space to the five nearest training structures, thresholded at the 95th percentile of training leave-self-out distances; this is the k-nearest-neighbour formulation among the standard families of domain definition [48, 49]. The choice of metric is not neutral for binary fingerprints, since Euclidean distance counts shared absences as evidence of similarity while Tanimoto does not [50]; we report both and the resulting domain memberships in SI §S10. Conformal intervals have themselves been proposed as a domain criterion [51], and out-of-distribution alerting serves the same function in open-set classification settings [52, 53]. A structure was *cleared* when the lower bound of its 90% interval exceeded the hazard threshold. Because the endpoint constructions differ in protectiveness, clearance was evaluated across a continuum of thresholds rather than at a single value, and reported at three reference points: 0.1 mg/L, the value used with the protective minimum; 1 mg/L, the Globally Harmonised System acute aquatic Category 1 boundary; and a stringency-matched value obtained by shifting 0.1 mg/L by the measured offset between the two endpoints.

2.6. Subclass assignment

PFAS were assigned to structure categories following the OECD (2021) definition and the 2018 OECD/UNEP structure series [54, 55], the same categorisation logic that underpins regulatory category and read-across approaches for the class [56]. The definitional gate requires at least one fully fluorinated, saturated methyl or methylene carbon with no hydrogen, chlorine, bromine or iodine attached; methine (>CF–) and unsaturated (=CF_2) carbons do not satisfy it. Categories were then assigned by parsing the molecule as C_nF_{2n+1}–X–R, with head-group identity taking precedence over incidental fluorination, so that a perfluoroalkyl ether acid is recorded as an ether acid rather than as a chain compound.

3. Results and discussion

3.1. The protective minimum tracks how much a substance has been tested

Across the 87 parent structures with freshwater records, the protective minimum correlated with the logarithm of the number of available records at $\rho = -0.749$ ($p = 7\times10^{-17}$; Figure 1a). The same correlation computed on the number of distinct tests rather than records is $\rho = -0.703$. The gradient is not subtle. Binned by record-count quartile, the median protective value falls monotonically from +1.16 $\log_{10}$ mg/L in the least-tested quartile through −0.52 and −2.00 to −3.76 in the most-tested, a difference of 4.9 $\log_{10}$ units (Mann–Whitney $p = 1\times10^{-7}$); the comparison between substances with three or fewer records and those with at least a hundred is 6.3 units. Either span is larger than the entire range the modelling literature attempts to predict.

The same substances illustrate the mechanism individually. Perfluorooctanesulfonic acid carries 1,548 distinct freshwater records from 690 tests and perfluorooctanoic acid 1,232 from 679; the minimum of each is −6.00. Under the protective convention each is summarised by the lowest value in its own large sample. These extreme values are not data-entry errors — they are genuine entries in a public, curated database — but they are the minima of very large draws from very wide distributions, and a substance tested twice cannot produce a comparable minimum however toxic it is.

3.2. Sampling depth accounts for 40–48% of the association

Correlation between effort and the protective value admits explanations that are not mutually exclusive. Substances of greater concern attract more

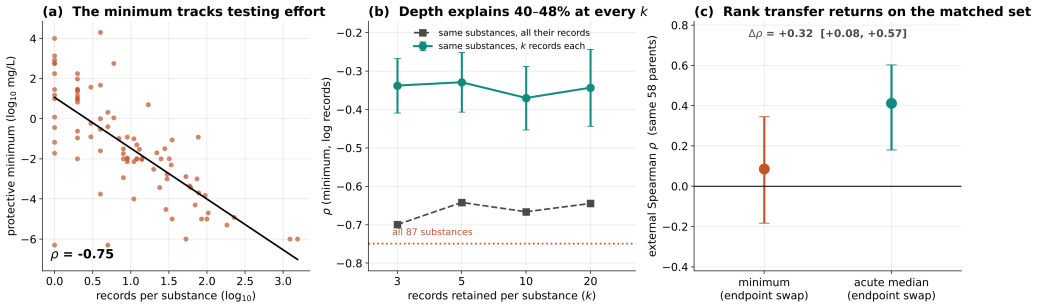


Figure 1: The protective endpoint is confounded with testing effort. (a) Protective minimum against the number of de-duplicated freshwater records per PFAS parent structure ($n = 87$). (b) Spearman correlation between the minimum and the record count after subsampling every eligible substance to depth k (teal), against the same substances using all of their records (grey); the gap between the two lines is the depth effect, and the dotted line is the value over all 87 substances. Bars are standard deviations over 400 draws. (c) External rank transfer of a fixed broad non-PFAS model on the 58 parent structures common to both endpoints, with 95% bootstrap intervals and the paired difference annotated.

testing, which would make the association causal in the reverse direction; more-tested substances also accumulate a wider range of species, durations and effect types, so their records are drawn from a more dispersed distribution; and the minimum of a larger sample is lower by construction. Subsampling isolates the last of these. Because a substance is eligible for depth k only if it has at least k records, the comparison must be made against the same substances at full depth rather than against the whole set, or the change in sample is read as an effect of depth. Against that baseline the depth effect was 0.36, 0.31, 0.30 and 0.30 at $k = 3, 5, 10$ and 20 (Figure 1b), i.e. 40–48% of the full-sample association and stable in k; the apparent anomaly at $k = 20$ in a whole-set comparison is an artefact of the shifting denominator.

We do not attribute the residual to reverse causation. That the measured toxicity of a PFAS varies substantially across studies, species and designs is itself well documented, and has been quantified by meta-regression across the class [57]. Within-substance dispersion itself rises steeply with effort in these data (SI §S3), and subsampling draws from each substance's own pool, so it preserves that substance's dispersion and assay mix by construction and cannot remove that channel. The residual $\rho \approx -0.34$ is therefore not attributable to sampling depth alone, and reverse causation and effort-dependent assay breadth are not separable in this design.

The effect is not peculiar to PFAS. Over all 4,757 chemicals with freshwa-

ter acute data, the acute minimum carried $\rho = -0.46$ against record count while the acute median carried $\rho = -0.16$ — a threefold difference between two summaries of identical records. The confound is a property of taking a minimum, not of the chemical class.

3.3. Transfer failure follows the confound, not the chemistry

A structure-based model has no access to a substance's testing history. If the protective endpoint is partly a measure of that history, the model should fail on it for reasons unrelated to chemistry. Holding the published broad non-PFAS model fixed and changing only the external target tests this directly (Figure 1c). An acute median is defined only for substances carrying acute records, so the four constructions are not available on identical substance sets; we therefore report the comparison on the 58 parent structures common to the protective minimum and the acute median. On those same 58 structures the minimum gave $\rho = +0.086$ (95% CI $[-0.18, +0.35]$) and the acute median $\rho = +0.412$ ($[+0.18, +0.60]$); a paired bootstrap on the difference gives $+0.321$ ($[+0.08, +0.57]$, $p = 0.005$). Only the acute median's interval excludes zero. Across all four constructions on their full available sets, the ordering of external rank performance is the ordering of freedom from the effort confound (SI Table S7).

We lead with rank rather than R^2 deliberately. The protective minimum sits 2.2 $\log_{10}$ units below the training target's mean, and that constant accounts for most of its apparent R^2 collapse: removing a single intercept moves the minimum from $R^2 = -0.714$ to $+0.011$ and the acute median from $+0.247$ to $+0.325$. The R^2 contrast therefore restates the location shift that the order-statistic mechanism produces, and is not independent evidence about predictive structure. The rank contrast is, and it survives the matched comparison above. Variance compression is also visible and should be reported: the ratio of predicted to observed standard deviation is 0.34 on the minimum against 0.59 on the acute median.

This also accounts for an observation reported previously without a mechanism. Restricting to acute endpoints has been noted to improve external rank correlation, and attributed to heterogeneity in assay availability. The present analysis identifies what that heterogeneity is: acute-only restriction removes most of the effort confound, because acute lethality endpoints are reported in a narrower range and the median is not an extreme order statistic.

3.4. Consistent rebuilding restores predictive validity

The comparison above changed the external target while leaving the training target as published. Rebuilding both consistently on the acute median gave a broad training set of 3,718 structure-locked non-PFAS chemicals, of which 3,717 yielded a valid fingerprint and entered the model, and an external set of 58 structure-locked PFAS. Broad cross-validation improved from $R^2 = 0.387 \pm 0.024$ to 0.443 ± 0.029. External validation improved from $R^2 = -0.580$ to $+0.366$ and from $\rho = -0.071$ to $+0.460$ (95% CI $[+0.23, +0.65]$) (Figure 2a).

Interval behaviour improved correspondingly, with two qualifications we state rather than leave to the reader. Marginal calibration, which undercovered PFAS badly on the protective endpoint (0.659 at nominal 0.90), reached 0.851, and PFAS-conditional calibration reached 0.897. Both are estimated on 58 and 29 external structures respectively, so their intervals are wide: 95% CI $[0.73, 0.93]$ for the marginal and $[0.73, 0.98]$ for the conditional. Neither differs significantly from the nominal 0.90, but the design could not have detected coverage as low as 0.81 either, and these figures should be read as estimates with a spread of about ± 0.10, not as demonstrations of validity (SI Table S9). Marginal calibration moreover splits the broad non-PFAS set and tests on PFAS, so exchangeability fails by construction and no conformal guarantee applies to it; the conditional procedure, calibrated and tested within PFAS, is the one with a theoretical basis, and it is also the one we deploy.

The conditional median 90% interval narrowed from 10.53 to 4.48 $\log_{10}$ mg/L (2.5th–97.5th percentile over 40 calibration splits $[3.34, 5.35]$; Figure 2b), but the greater part of that narrowing is the target being less dispersed rather than the model being better. On the 58 matched structures the ratio of standard deviations is $1.58/2.67 = 0.59$, so a width of about 6.2 units would follow from the change of target alone; the achieved 4.48 improves on that, and the comparison that carries the practical meaning is 4.48 against the 2.43 that would be required for the median in-domain structure to clear at the 1 mg/L decision point (SI §S6). We therefore do not claim that the interval was halved by better prediction.

3.5. Clearance still fails once the threshold is matched

Applied to the untested inventory, the rebuilt model placed 16,070 of 17,619 unique structures inside the applicability domain with a median interval width of 4.87 $\log_{10}$ units, and cleared 5,441 of them (33.9%) at the

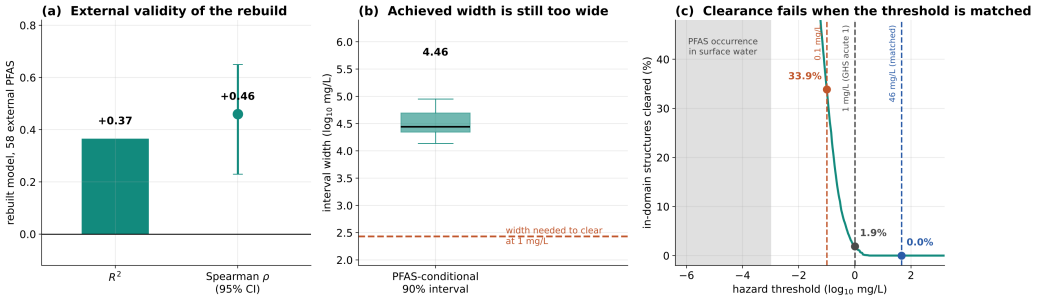


Figure 2: The rebuilt model. (a) External validation on 58 structure-locked PFAS, with the 95% bootstrap interval on Spearman ρ. (b) PFAS-conditional 90% interval width over 40 calibration splits, against the width that would be required for the median in-domain structure to clear at 1 mg/L. (c) Fraction of in-domain inventory structures cleared as a function of the hazard threshold. Three reference thresholds are marked: the value used with the protective endpoint, the GHS acute Category 1 boundary, and one construction of a stringency-matched value. The shaded band is the concentration range in which PFAS are reported in surface water; the modelled effect concentrations lie several orders of magnitude above it, as hazard terms necessarily do.

0.1 mg/L threshold used with the protective endpoint. That number must not be compared with the zero reported for the protective endpoint, and the reason is quantitative. On the same 58 structures the acute median lies above the protective minimum by 2.66 $\log_{10}$ units in the mean and 2.32 in the median, and 93.1% of measured PFAS exceed 0.1 mg/L on the acute median against 39.7% on the minimum. The threshold means something different on each endpoint, and holding it fixed while changing the endpoint relaxes the decision rather than improving it.

Evaluating clearance across thresholds removes the ambiguity (Figure 2c). At the GHS acute aquatic Category 1 boundary of 1 mg/L, 306 structures clear (1.9%). At a threshold shifted to match the protective endpoint for stringency, none clear. That threshold is a construction, not a regulatory object, and it is not unique: the mean paired offset puts it at 46 mg/L, the median offset at 21 mg/L, and matching instead on the fraction of measured PFAS falling above the cut-off (39.7% on the minimum) puts it at 41 mg/L. Clearance is zero at all three, and at every threshold at or above 10 mg/L, so the conclusion does not depend on which construction is chosen (SI Table S10). The published conclusion that no untested PFAS can be cleared therefore survives, and survives on a model that now has demonstrated external validity rather than on one that had none.

What changes is the explanation, and the two reference thresholds fail for different reasons. The obstacle is not that structure-based models cannot predict PFAS aquatic toxicity; on a properly constructed endpoint they predict it with $R^2 = +0.37$ and a rank correlation whose interval excludes zero. At the GHS Category 1 boundary the obstacle is uncertainty: the median in-domain structure would clear with a 90% interval of 2.43 $\log_{10}$ units against the 4.87 achieved, so halving the interval again would be sufficient. That is a concrete target, and one the previous framing could not have specified, because an endpoint confounded with testing effort does not define a stable distance to any threshold.

At the stringency-matched threshold the obstacle is different and more fundamental. There the median in-domain structure has a point prediction *below* the threshold, so the required half-width is negative and no narrowing of the interval would clear it. At protective stringency the model is not uncertain about whether these structures are low hazard; it predicts that most of them are not. Separating these two regimes matters for research planning, because only the first is addressed by better uncertainty quantification.

3.6. Structure classes cannot be resolved at available sample sizes

A natural next question is whether transfer differs across PFAS structure classes, since the chemical argument for extrapolation failure is class-specific, and aquatic potency within the class does vary systematically with chain length and head group [58, 59, 60]. We tested this on OECD (2021) categories and report a negative result with its power requirement, because the point estimates invite over-reading.

We classify on the acute median, the endpoint this paper endorses: a class-resolved analysis of the protective minimum would inherit the effort confound a class at a time. Of the 87 parent structures, the OECD definitional gate admits 83; the four it excludes are difluoroacetic acid, whose CHF_2 carbon is not fully fluorinated, pentafluoroaniline, pentafluorobenzaldehyde and 1,1,2-trichlorotrifluoroethane. Fifty-six of the 83 carry acute records and enter the comparison, distributed across nine categories, of which six reach five substances; the two smallest, perfluoroalkyl ether acids and ether others at three substances each, and perfluoroalkyl phosphorus compounds, which carry no acute record at all, are reported in SI Table S11 but not estimated, since a bootstrap interval on three points is degenerate.

No category resolves. Pooled across all classified PFAS the model transfers at $\rho = +0.41$ (95% CI $[+0.18, +0.61]$, $n = 56$), but every one of the six

estimable categories has an interval covering zero, from $\rho = -0.44$ for fluorotelomers ($n = 9$) to $+0.68$ for the residual perfluoroalkyl category ($n = 7$) (Figure 3). This is a cleaner negative result than the one we reported previously on a superseded data extraction, where a single category appeared to exclude zero; that estimate does not survive recomputation on the present set, and we note the correction explicitly. We also tested whether class-level transfer merely tracks class-level testing effort, which would make the figure a third measurement of the confound rather than a statement about chemistry: it does not ($\rho = -0.49$, $p = 0.33$, between class rank correlation and class mean $\log_{10}$ record count across the six estimable classes — negative, and far from significant on six points).

Power is the binding constraint. At these category sizes, 80% power at $\alpha = 0.05$ detects only $|\rho| \geq 0.71$ to 0.92. Resolving differential transfer at a more realistic effect size of $\rho = 0.3$ would require approximately 85 measured substances per category, or 129 with correction for six comparisons — roughly 510 to 780 measured PFAS against the 87 parent structures the public record currently supports. This is the quantity a programme that wants class-resolved screening would need to measure, and it is six to nine times what exists. Measuring that many substances at random is not the only option: where the cost of an experiment dominates, the selection of which substances to test is itself a design variable, and active or model-guided selection routinely attains a given accuracy with a fraction of the measurements required under random sampling [61, 62, 63, 64]. We note it as the obvious route to the sample sizes above rather than as a result of this study.

3.7. Implications for screening practice

Four consequences follow for programmes that build or consume structure-based hazard screens.

First, the response variable deserves the scrutiny usually reserved for the model. A protective minimum taken over a heterogeneous record is not a property of the substance alone, and the degree to which it is not can be measured directly with the correlation reported here. We would suggest that any screening study report the association between its endpoint and the number of underlying records, as a routine diagnostic, in the way applicability domains and condition-dependence are now routinely reported [65].

Second, a model that fails external validation on a protective endpoint has not thereby been shown to lack chemical validity. The failure is consistent

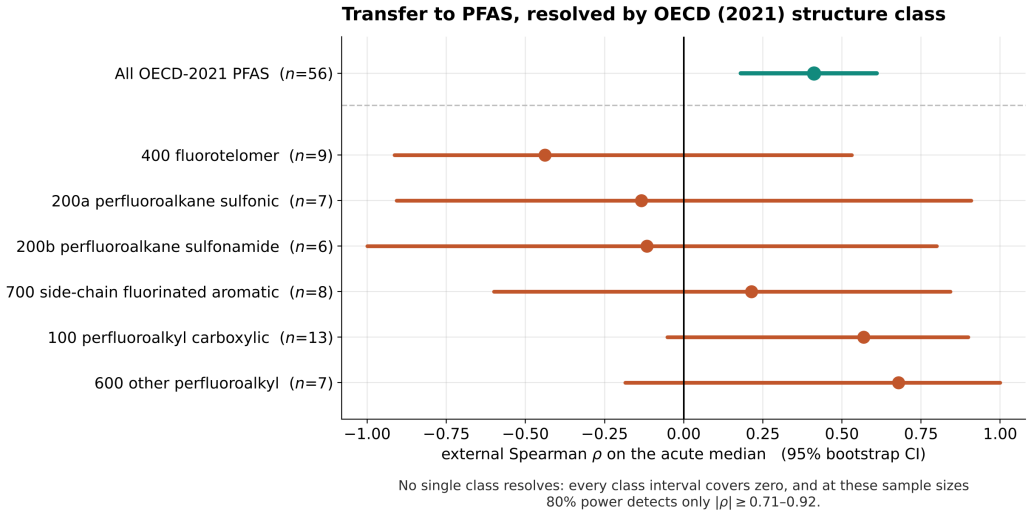


Figure 3: External rank correlation by OECD (2021) structure class on the acute median, with 95% bootstrap intervals. The pooled estimate excludes zero; no individual category does. Categories with fewer than five substances are listed in SI Table S11 and not estimated.

with the model being unable to predict testing history, which is not a chemical deficiency. Endpoint construction is not the only lever: on the earlier extraction of this work, holding the endpoint at the protective minimum and replacing the binary ECFP4 representation with 165 physicochemical descriptors also recovered rank transfer, from $\rho = -0.03$ to $+0.42$ (SI Table S12), and a ridge baseline outperformed the random forest on the same target (SI Table S13). Both tables are from the superseded extraction and we have not repeated them on the present one, but the direction is clear enough that the claim must be made in the weaker form. There is a mechanistic reason to expect representation to matter for this class: much of PFAS bioactivity is mediated by binding to serum proteins and to nuclear receptors, with potency depending on chain length and head group in ways a topological fingerprint does not encode [66, 67, 68]. Endpoint and representation therefore both matter. We report the endpoint comparison because it is the larger of the two on matched samples, because it is the one a screening programme controls when it defines its response variable, and because it is inexpensive; we do not claim that representation is irrelevant.

Third, the confound propagates to the concentrations at which PFAS are actually regulated, which are far below those modelled here. PFAS occur in

surface water at nanograms to low micrograms per litre; the European environmental quality standard for perfluorooctanesulfonate in inland surface water is 0.65 ng/L [69] and the United States maximum contaminant level for perfluorooctanoate and perfluorooctanesulfonate is 4 ng/L [70], while the effect concentrations modelled here, and the 0.1 to 1 mg/L decision boundaries applied to them, sit five to seven orders of magnitude higher. That gap is not an artefact of this study: it is the standard structure of hazard assessment, in which a predicted no-effect concentration is an effect concentration divided by an assessment factor, so the hazard term is necessarily derived at effect concentrations and only then brought down to the occurrence range. The consequence is that an effort confound in the effect term is not diluted by that division — an assessment factor is a multiplicative constant and leaves a 4.9 $\log_{10}$ unit gradient intact. A criterion derived at nanograms per litre from a minimum-based hazard term inherits the whole of it. This is why we regard the diagnostic as environmentally consequential rather than merely methodological, and it is also why we do not present the clearance fractions as risk estimates: a clearance rule that compares a predicted effect concentration directly with a fixed cut-off, as the screening literature we re-examine does, carries no exposure term, and the fractions below should be read as a property of that decision rule.

Fourth, the protective convention is not thereby wrong. It remains the correct posture for a regulatory decision, because the lowest observed effect does bound the hazard. What the present results argue is that it is unsuitable as a *regression target*, and that screening pipelines should model an effort-stable quantity and apply protectiveness at the decision step rather than embedding it in the response variable where it cannot be separated from testing history.

We are deliberate about what that does and does not propose. It is not a proposal to substitute a median into an existing assessment-factor schedule. Those schedules are calibrated over a *lowest* value from a specified set of trophic levels — ECHA Guidance R.10 assigns a factor of 1000 to the lowest short-term L(E)C50 across three trophic levels [71] — and no published calibration exists for a median, so swapping the summary underneath an unchanged factor would leave the factor undefined. The claim is narrower: when a structure-based model is fitted in order to *prioritise* untested substances, its regression target should be effort-stable, and protectiveness should re-enter when the prioritised substances are carried into an assessment. We also note what the record-level acute median is not. It is a median

over records, not over species: the ECOTOX extraction used here does not carry a species field, so a substance tested fifty times in *Daphnia magna* and once in a fish is summarised by a value that is effectively the *Daphnia* result. The species-aggregated alternatives — a geometric mean per species feeding a species sensitivity distribution, with the HC5 as the summary [72] — are both effort-stable and protective, have a mature methodological literature, and are the natural target for a screening model built on a species-resolved extraction. We did not evaluate them here because the extraction we re-used from the study under re-examination does not carry species, and we regard that comparison as the most useful next step rather than a gap we can close in this paper.

3.8. Limitations

The external PFAS set contains 58 structure-locked substances on the acute median, smaller than the 87 parent structures available on the protective endpoint, because restricting to acute endpoints removes the 29 substances whose records are entirely chronic. This is the principal limitation and it bounds the precision of every external statistic reported. Structures resolved for 90.6% of chemicals; the unresolved remainder may be non-random with respect to chemical class, which we could not assess. The acute median remains a mixture over species, exposure duration and effect type, and is effort-stable rather than effort-free. The clearance analysis evaluates aquatic effect concentrations only and says nothing about exposure, persistence or bioaccumulation, which are the properties that in practice drive PFAS regulatory action; a substance with a high acute LC50 and a decades-long half-life is not low concern, and no aquatic-toxicity model of any quality would establish that it was. Finally, the residual association that survives depth equalisation is not identified: this design cannot separate reverse causation from effort-dependent assay breadth, and the latter is not removed by subsampling.

4. Conclusions

Summarising a substance by the lowest effect concentration ever reported for it makes the resulting value depend on how often it has been tested. Across 87 PFAS parent structures the protective minimum correlated with record count at $\rho = -0.749$, and equalising sampling depth against the eligible-subset baseline identified 40–48% of that as an extreme-order-statistic artefact; across 4,757 chemicals the acute minimum carried three times the

confound of the acute median. Because a structure-based model cannot predict testing history, this confound contributes to the external validation failures reported for PFAS screening: holding a published model fixed and changing only how the same records are summarised raised rank transfer on the same 58 parent structures from $\rho = +0.09$ to $+0.41$, a paired difference of $+0.32$ (95% CI $[+0.08, +0.57]$, $p = 0.005$).

Rebuilding training and external sets consistently on the acute median restored external validity ($R^2 = +0.37$; $\rho = +0.46$, 95% CI $[+0.23, +0.65]$) and narrowed the PFAS-conditional interval to 4.48 $\log_{10}$ mg/L, most of that narrowing being attributable to the less dispersed target rather than to better prediction. Clearance nevertheless failed at a stringency-matched threshold, where no in-domain inventory structure cleared, and at the GHS acute Category 1 boundary, where 1.9% did. The conclusion that structure-based screening cannot presently clear untested PFAS therefore stands, but it stands for a reason the earlier framing did not identify: not that such models cannot predict PFAS aquatic toxicity, but that a validated model's interval, and at protective stringency its point prediction, remain on the wrong side of the decision threshold. The practical recommendation is to model an effort-stable endpoint, report the endpoint–effort association as a routine diagnostic, and apply protectiveness at the decision step rather than inside the response variable.

CRediT authorship contribution statement

Jeongyun Kim: Methodology, Software, Formal analysis, Writing – original draft. **Jin Yoo:** Data curation, Validation, Formal analysis, Writing – review & editing. **Ku Kang:** Conceptualization, Supervision, Methodology, Writing – original draft, Writing – review & editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

During the preparation of this work the authors used a large language model to assist with code development for data extraction and analysis, and with language editing. The authors reviewed and edited all output and take full responsibility for the content of the publication.

Data availability

All extraction and analysis scripts, the structure-locked tables, and the inventory-level predictions are openly available at `https://github.com/bisu9082/pfas-clearance-validation`, tagged as release `v3.2`. Every number reported in this paper and its Supporting Information, and every figure including the graphical abstract, is regenerated by the scripts in `scripts/` from the inputs in `canonical_results/`; the correspondence between script, file and table is given in SI §S11. Superseded result files from the earlier version of this study are retained under `archive_v2/` and are clearly marked as such; they are the source of SI Tables S12–S14 and of no other table. The underlying source data are public: the U.S. EPA ECOTOX Knowledgebase ASCII release of 15 September 2026, the CompTox Chemicals Dashboard PFASSTRUCTV6 list, and PubChem for structure resolution. Raw third-party databases are not redistributed; the repository documents the exact release used.

Acknowledgements

The authors thank the teams maintaining the U.S. EPA ECOTOX Knowledgebase, the CompTox Chemicals Dashboard and PubChem.

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